

Instructor: Date:

Learner: Regional Branch:

Program: Center:

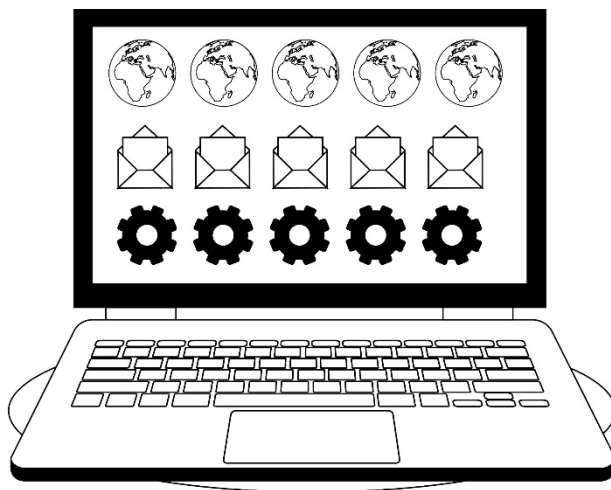
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Let's Start!

Organizing Processes: The Concept of Queue



Learning Outcome:

Understand key information about the concepts, principles, and methodologies in simple spoken and written situations. Talk about programming techniques, use cases, kinds, and sequences in simple spoken and written situations.

Can Do Statements:

- I can explain what queue is in software programming.
- I can talk about concepts related to queue and software programming.
- I can understand different information about the concept of queue.

Ask your partner:

- Why do computers become slow?
- How do computers run programs?



Let's Explore!

Kyle and Dave review their presentation for the programming class



Instructions: Read the text below.

- Kyle:** To get a good grade in the programming class, we should practice the presentation we prepared about the concept of queue.
- Dave:** Talking about the concept of queue is not very difficult. I'm sure we'll get a good grade. However, if you want to practice, I will ask you some questions.
- Kyle:** That sounds great. I'm ready.
- Dave:** Can you explain to me what queue is?
- Kyle:** Well, to understand the concept of queue, we just have to imagine a list of arranged items, and the organization of these items depends on their priority. Computers often carry out different processes at the same time. However, the operating system needs to prioritize to avoid delays or errors.
- Dave:** Excellent, now, what concepts should we take into account when we talk about queue?
- Kyle:** To be able to work with queue, it's important to take into account concepts such as FIFO which means first in, first out; buffering which refers to storing data temporarily for later use, and finally, you need to know that the name of the place where we add items is the tail or rear, and the name of the place where we remove items is the head or front.
- Dave:** Having those concepts clear helps a lot!
- Kyle:** Now, you should close the presentation by explaining what you understood about the concept of queue.
- Dave:** So, to sum up, I understood that a queue organizes different items by priority, and the system adds these items at the tail and removes them from the head, for this reason, we can say that there is a FIFO structure. The main purpose of a queue is organizing processes and prioritizing applications that should get the attention of the processor first. What else should we say?
- Kyle:** To create a queue, the system finds a block of memory and sets up pointers, and that we can find straight queues or circular queues.
- Dave:** Oh, that's right!



Let's Practice!

Different Ideas to Understand the Concept of Queue

Instructions: Write the correct answer on the line provided. Read the sentences aloud. Check your pronunciation with your instructor.

1. The system needs to organize processes, _____, it uses queues.
 - a. for reason
 - b. however
 - c. for this reason
 - d. how ever
2. The processor needs to perform all tasks, _____, it needs to prioritize those which are more important.
 - a. for reason
 - b. however
 - c. for this reason
 - d. how ever
3. The system needs to create queues, _____, it finds blocks of memory to set up pointers.
 - a. for reason
 - b. however
 - c. for this reason
 - d. how ever
4. The system needs to remove the tasks that arrive first, _____, it uses the FIFO structure.
 - a. for reason
 - b. however
 - c. for this reason
 - d. how ever
5. We can use queue to organize information, _____. We don't know whether to use a circular or a straight queue.
 - a. for reason
 - b. however
 - c. for this reason
 - d. how ever
6. The first item to get in is the first one to get out, _____, the system prioritizes the information first.
 - a. for reason
 - b. however
 - c. for this reason
 - d. how ever



Let's Practice!

Queues and System Programming: Efficient Storage*

*"Storage" relates to space to place, keep and save elements.

Instructions: Read the text. Underline all the correct answers. Check your answers with your partner. Discuss.

Programming needs procedures to store information. For this reason, an operating system needs to prioritize data to avoid delays on the processes. Queues are essential for organizing elements in software programming.

1. Which of these sentences are true?
 - a. Programming needs procedures to store information.
 - b. Unlimited storage of data is known as buffering.
 - c. Prioritizing data to avoid delays on the processes is crucial for an operating system.
 - d. Prioritizing information is a waste of time.
 - e. Organizing elements in software programming is possible thanks to queues.



Extension Activity

Let's define what a queue is.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Investigate about queues in software programming.
- Create a list of examples of processes that involve queues.
- Share the list with your class, remember to use statements using for this reason and however.

Self-Assessment

Yes

No

- I understand what a queue in software programming is.
- I list examples of queues.
- I share my work with my classmates.

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Target Words

- **arranged** (adjective): to be organized or placed
e.g. We just have to imagine a list of arranged items, and the organization of these items depends on their priority.
- **delay** (noun): a postponement
e.g. The operating system needs to prioritize to avoid delays or errors.
- **buffering** (noun): referring to temporary storage
e.g. It's important to take into account concepts such as FIFO which means first in, first out; buffering which refers to storing data temporarily for later use.
- **prioritize** (verb): to put in first place and give priority
e.g. The operating system needs to prioritize to avoid delays or errors.
- **carry out** (phrasal verb): to make or perform an action or plan
e.g. Computers often carry out different processes at the same time.
- **sum up** (phrasal verb): to express an idea briefly and concisely
e.g. So, to sum up, I understood that a queue organizes different items by priority.



Learning Points

Connectors for Contrast and Addition: For This Reason and However

There are different tools we can use to improve our writing skills, for example, the connectors **for this reason** and **however**.

The connector **for this reason** explains the cause of something. First, you give the explanation and then connect it to the cause or final result of it.

For example:

He had a test on Monday, for this reason he studied very hard on weekend.

The connector **however** makes a contrast between two opposite ideas that are connected.

For example:

He studied very hard for the test, however, he forgot some information.

Examples

- *Computers often carry out different processes at the same time. However, the operating system needs to prioritize to avoid delays or errors.*
- *I understood that a queue organizes different items by priority, and the systems adds these items at the tail and removes them from the head, for this reason, we can say that there is a FIFO structure.*

Infinitives to Start Sentences and Gerunds as Nouns

We can start sentences using **Infinitives** and **gerunds**, they have different uses.

Infinitives are verbs with the preposition **to** at the beginning. We can use the **infinitives** to begin a sentence. Usually these sentences have two clauses. The first introducing the idea linked to the infinitive, and the second making an extension of the idea.

Example: **To discard** an item, the system always takes it from the head of the queue.

In the case of **gerunds**, we can identify them because they finish with **-ing**. In this case, the gerunds become a noun to express the idea that defines the verb.

Example: **Programming** is very difficult for some people.





Learning Points

Examples

- *So, to prioritize what application should get the attention of the processor, the operating system organizes information in a queue.*
- *Organizing processes and prioritizing applications should get the attention of the processor first.*

Examples

- *Talking about the concept of queue is not very difficult. I'm sure we'll get a good grade. However, if you want to practice, I will ask you some questions.*
- *Talking about the concept of queue is not very difficult.*
- *Having those concepts clear helps a lot!*

List of Programming and Computer Science Terms

Let's take a look at some computer programming terminology.

- **Abstract Base Class:** In object-oriented programming, the base class from which other classes derive.
- **Abstract Type:** A nominative type system declared by a programmer.
- **Acrobat:** A document exchange software belonging to Adobe Systems which allows to convert documents into PDF.
- **ActionScript:** A scripting language similar to JavaScript.
- **ActiveMovie:** A multimedia streaming technology that allows users to watch multimedia content.